A Frustrated-Conditioning Obstruction for Local Yang–Mills Defect Domination

Why dependent ultraviolet control must charge the boundary to a contour

Mathematical research dossier

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Abstract

An ultraviolet one-plaquette heat-kernel tail can be small while a worst-case conditional tail in the interacting link measure is large. We prove this separation exactly on the side-two four-dimensional SU(2) block. An interior link belongs to six plaquettes. The center-staple map from the remaining links onto those six plaquettes has $\mathbb{Z}_2$-rank six, so an admissible center-valued conditioning can give five identity staples and one negative-identity staple. The selected link then has conditional density

$$\gamma_t(\,\mathrm{d}U) \propto p_t(U)^5 p_t(-U)\,\mathrm{d}U.$$

Its weak-coupling class-angle rate function is

$$I_6(\theta) = 5\theta^2 + (\pi-\theta)^2.$$

The unique minimizer is $\theta_* = \pi/6$, leaving the designated frustrated plaquette at angle $5\pi/6$. If its good-curvature radius is $0 < \rho < 5\pi/6$, then

$$\lim_{t\downarrow 0} 2t \log \gamma_t\{\theta_{\mathrm{target}} < \rho\} = -6\left(\frac{5\pi}{6} - \rho\right)^2.$$

Thus the conditional bad probability tends to one. In particular, no rare-defect estimate uniform over arbitrary microscopic boundary conditionings can initialize the Stage 09 independent-polymer benchmark.

This obstruction eliminates a proof mechanism, not the constructive programme. A viable dependent estimate must retain the cost of the frustrating boundary: condition on a good collar, integrate a multi-link block, or charge the entire connected frustration to a labeled contour that also records center and topological data.

Contents

1 The dependence question **2**

2 Exact center-staple realization **2**

3 The exact one-link conditional law **3**

4 Small-time rate theorem **3**

5 Executable calibration **4**

6 What proof route has been eliminated? **5**

7 Replacement: charged and typed contours **5**

8 Frontier audit **5**

1 The dependence question

Let B be the cubical complex with vertices $\{0,1,2\}^4$. It has 216 links and 216 plaquettes. With heat generator $\frac{1}{2}\Delta_{\mathrm{SU}(2)}$, the regulated pure plaquette weight is

$$\prod_{p\in B_2} p_t(U_p) \prod_{e\in B_1} \mathrm{d}U_e.$$

Stage 09 proved that at $t = 3\cdot 10^{-6}$ a local nonlinear action chart and a standalone one-plaquette tail can satisfy the desired inequalities at the same scale. The missing step was dependence: could the standalone tail be replaced by a bound valid under the link interactions?

A tempting sufficient condition would be

$$\sup_{\omega} \Pr_t\{U_p \text{ leaves the radius-}\rho \text{ chart} \mid \omega\} \le \varepsilon(t), \qquad \varepsilon(t) \longrightarrow 0, \tag{1}$$

where ω ranges over all surrounding microscopic link conditionings. Such a bound could feed a Dobrushin or stochastic-domination argument without tracking the probability of ω. The central result shows that (1) has the wrong quantifier.

Theorem 1.1 (Frustrated-conditioning obstruction)**.** *On the open side-two four-dimensional* SU(2) *block, there is an interior link e and a center-valued assignment of every other link such that five plaquette staples incident to e equal I and one equals $-I$. Conditioned on those other links, U_e has density*

$$\gamma_t(\,\mathrm{d}U) = Z_t^{-1} p_t(U)^5 p_t(-U)\,\mathrm{d}U. \tag{2}$$

If the plaquette with staple $-I$ is declared good when its class angle is less than ρ, then for every $0 < \rho < 5\pi/6$,

$$\lim_{t\downarrow 0} 2t \log \gamma_t\{\theta_{\mathrm{target}} < \rho\} = -6\left(\frac{5\pi}{6} - \rho\right)^2 < 0. \tag{3}$$

Consequently

$$\lim_{t\downarrow 0} \sup_{\omega} \Pr_t\{U_p \text{ is bad} \mid \omega\} = 1.$$

At the Stage 09 curvature radius $\rho = 0.0082$, the rate gap in (3) is 40.8661445. The obstruction is therefore separated by a large action-scale margin, not a borderline numerical constant.

2 Exact center-staple realization

Choose a link of maximum plaquette degree. The exact incidence matrix of the block has six nonzero entries in its column. Hold the selected link variable and assign every other link a center sign $(-I)^{x_e}$, $x_e \in \mathbb{Z}_2$. Orientation disappears because $(-I)^{-1} = -I$. The six staple signs are therefore a linear map

$$C : \mathbb{Z}_2^{215} \longrightarrow \mathbb{Z}_2^6$$

obtained by deleting the selected incidence column and reducing the remaining six rows modulo two.

Proposition 2.1 (Staple surjectivity)**.** *The map C has rank six. In particular, every prescribed pattern of center staples on the six incident plaquettes is realized by an assignment of the remaining links.*

Proof. For each incident plaquette, take the link opposite and parallel to the selected link. These six opposite links are distinct. The opposite link for a given plaquette belongs to that incident plaquette and to none of the other five incident plaquettes. The six corresponding columns of C therefore form the 6×6 identity matrix after row ordering. Hence $\operatorname{rank}_{\mathbb{Z}_2} C = 6$. □

The executable certificate independently constructs the full 6×215 matrix, performs Gaussian elimination over $\mathbb{Z}_2$, and solves for a vector with exactly one requested negative staple. It selects link

$$e = ((0, 1, 1, 1), 0),$$

whose six incident plaquettes have global indices $21, 44, 54, 58, 59, 60$. A solution uses one negative-center link and verifies the requested staple vector exactly.

Remark 2.2 (Positive-measure stability)**.** The exact center assignment is a legitimate value of the regular conditional density. Moreover, the six staples and the conditional density depend continuously on the surrounding links. The displaced conditional minimum and the bad-probability conclusion persist for a neighborhood of the center assignment. The obstruction is therefore not an artifact of selecting a single measure-zero boundary value.

3 The exact one-link conditional law

All plaquette factors not incident to e are constant under conditioning. For an incident identity staple, conjugation invariance and inversion symmetry give the factor $p_t(U_e)$. For the one negative-center staple the factor is $p_t(-U_e)$. This proves (2).

Write $\theta \in [0, \pi]$ for the class angle of U_e. The class angle of $-U_e$ is $\pi - \theta$. Radial Haar integration therefore gives

$$\gamma_t(\,\mathrm{d}\theta) \propto \sin^2\theta\, p_t(\theta)^5 p_t(\pi - \theta)\,\mathrm{d}\theta. \tag{4}$$

The target plaquette is good precisely when

$$\pi - \theta < \rho, \qquad \text{or equivalently} \qquad \theta > \pi - \rho.$$

This conditional law exposes why a one-plaquette marginal is insufficient. The five aligned plaquettes collectively compete with the designated one. Weak coupling does not force every factor independently to its own identity; it minimizes their total action.

4 Small-time rate theorem

The SU(2) heat kernel in the declared normalization has the class-angle small-time rate

$$\lim_{t \downarrow 0} -2t \log p_t(\theta) = \theta^2, \qquad 0 \le \theta \le \pi. \tag{5}$$

The exact Poisson-image representation proves this directly: the shortest geodesic image supplies the exponential rate, while all prefactors and the cut-locus multiplicity contribute only subexponential powers of t.

Proposition 4.1 (Conditional large-deviation rate)**.** *The probability measures* (4) *obey the class-angle rate function*

$$I_6(\theta) = 5\theta^2 + (\pi - \theta)^2, \tag{6}$$

up to subtraction of its minimum. The unique minimizer is

$$\theta_* = \frac{\pi}{6}, \qquad I_6(\theta_*) = \frac{5\pi^2}{6}.$$

The target plaquette angle at the minimizer is $5\pi/6$.

Proof. Apply (5) to the six heat-kernel factors in (4). The Haar factor $\sin^2\theta$ is subexponential and does not alter the rate. Completing the square gives

$$I_6(\theta) = 6\left(\theta - \frac{\pi}{6}\right)^2 + \frac{5\pi^2}{6},$$

which proves every assertion. □

Proof of Theorem 1.1. Proposition 2.1 gives the boundary assignment and Section 3 gives the conditional law. If $\rho < 5\pi/6$, the good region $[\pi - \rho, \pi]$ lies strictly to the right of θ_*. Its rate infimum is attained at $\pi - \rho$. The Laplace principle and Proposition 4.1 give

$$\begin{aligned} &I_6(\pi - \rho) - I_6(\theta_*) \\ &\qquad = 6\left(\pi - \rho - \frac{\pi}{6}\right)^2 = 6\left(\frac{5\pi}{6} - \rho\right)^2. \end{aligned}$$

This proves (3). The good probability tends to zero, so the bad probability tends to one. Taking the supremum over boundary conditionings can only increase it. □

5 Executable calibration

The implementation has two logically distinct parts. Exact finite algebra constructs the incidence restriction, computes its $\mathbb{Z}_2$ rank, solves the staple equation, and evaluates the closed rate formulas. Numerical calibration uses a signed-log Poisson-image sum and logarithmic Gauss–Legendre quadrature. It never exponentiates the ultraviolet probability and therefore remains stable when its logarithm is of order millions.

Table 1: Convergence of the conditional good-event logarithmic rate for $\rho = 0.0082$.

t	mode θ	$-2t \log P(\text{good})$	relative error
0.05	0.541692	39.578307	$3.151 \cdot 10^{-2}$
0.02	0.530832	40.274290	$1.448 \cdot 10^{-2}$
0.01	0.527454	40.570358	$7.238 \cdot 10^{-3}$
$3 \cdot 10^{-6}$	0.523599	40.866026	$2.908 \cdot 10^{-6}$

The exact limiting value is 40.866145. At the Stage 09 heat time the computed log good probability is approximately $-6.811004 \cdot 10^6$. Floating-point representation consequently rounds the bad probability to one, while the log-domain certificate preserves the quantitative separation. The numerical values calibrate the analytic theorem; they are not used to prove it.

6 What proof route has been eliminated?

The theorem rules out any dependent-defect argument whose decisive input is a uniform estimate over arbitrary microscopic exterior links, such as

$$\sup_{\omega} \Pr(B_p \mid \omega) \le p(t), \qquad p(t) \longrightarrow 0,$$

for $B_p = \{\theta_p \ge \rho\}$ with fixed $\rho < 5\pi/6$. It also rules out inserting the Stage 09 standalone tail into such an estimate: the standalone law and the frustrated conditional law have different rate minimizers.

The result does *not* show that bad blocks are frequent under the full unconditioned measure. The frustrating boundary may itself have a large action cost. Taking a supremum makes that cost disappear, which is precisely why the method fails. Nor does the result rule out a conditional theorem restricted to a good boundary collar.

7 Replacement: charged and typed contours

A viable activity expansion must include the cause of the frustration in the same connected object whose weight is being estimated. The minimal typed defect alphabet should distinguish

$$\begin{aligned}\mathsf{Defect} = {} & \mathsf{CurvatureExit}(\rho) \mid \mathsf{OrbitAtlasFailure}(c) \mid \mathsf{CenterFlux}(z) \\ & \mid \mathsf{BoundaryHolonomy}(h) \mid \mathsf{TopologicalCharge}(q) \mid \mathsf{InteractionNormFailure}(n).\end{aligned}$$

The labels cannot be collapsed to a single bad-block bit before their action and blocking rules have been charged.

The next theorem should combine two estimates with identical parameter registries. First, for a gauge-invariant good-collar event $G(A)$ surrounding a block B, prove

$$\sup_{\omega \text{ compatible with } G(A)} \Pr\{B \text{ exits the chart} \mid \omega\} \le p_{\mathrm{good}}(t, \rho, A). \tag{7}$$

Second, control failure of the collar by labeled activities:

$$\sup_{L} \sup_{o} \sum_{X \ni o} e^{\alpha|X|} |z_{\mathrm{collar},L}(X)| < \delta. \tag{8}$$

Alternatives that preserve the same information are full multi-link block integration and reflection-positive chessboard estimates. In each case the center-valued frustration must be paid for, not treated as free boundary data.

8 Frontier audit

Claim	Status	Scope
Degree-six interior-link incidence	proved	finite 2^4 block
$\mathbb{Z}_2$ center-staple surjectivity	proved	exact finite algebra
Conditional law (2)	proved	one admissible conditioning
Rate minimizer and gap	proved	$t \downarrow 0$ conditional law

Claim	Status	Scope
Worst-case uniform-tail obstruction	restricted	microscopic conditionings
Good-collar theorem (7)	open	interacting link measure
Charged activity sum (8)	open	uniform in volume
Repeated RG and basin entry	open	cutoff and scale uniformity
Continuum construction and Hamiltonian gap	open	Clay endpoint

The strongest defensible level is **C2, restricted obstruction theorem**. The result makes the dependent-measure target more precise by eliminating an over-strong quantifier. It does not supply the replacement activity expansion, construct continuum Yang–Mills theory, or prove a mass gap.

References

[1] P. Menotti and E. Onofri, "The action of $SU(N)$ lattice gauge theory in terms of the heat kernel on the group manifold," *Nuclear Physics B* **190** (1981), 288–300.

[2] T. Balaban, "Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions," *Communications in Mathematical Physics* **109** (1987), 249–301.